

From Finite Register Signatures to Physical Readability: Block Condensation, Retention, and a BBP-Related Limit Form

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Abstract

This paper introduces a finite register construction for reconstructing physical readability from continuable difference. Its central object is a block-condensed register signature that reproduces the coefficient and slot structure of the Bailey–Borwein–Plouffe series for π without inserting a physical constant. The result is treated as an internal limit signature of the register apparatus, not as a derivation or empirical identification of π .

The construction separates register multiplicity from role load. A triadic register $R_3 = 0\,111\,222\,3$ is obtained from the multiplicity structure $1, 3, 3, 1$ of three reference positions, their pair relations, and their triadic mediation. Role loads are assigned by a distinct rank-normalized rule $n \mapsto -n/3$, with drift-free closure at the initiating role. Block condensation, retention, and recursive projection then yield a BBP-related limit form at one extremal pole, while the opposite pole gives a distinct non- π residual form.

The signature is embedded in a complete finite path space of **140** admissible executions over the fixed inventory **1112223**. Inventory-preserving recombination on unordered path pairs defines the threshold at which stable structural roles can become projection classes. Physical readability is introduced only at this threshold: space, time, causality, field-like structure, particle-like units, and mass-like behaviour are treated as projection readings of stable structural modes rather than as primitives. Extended flavour-sector, astrophysical, fine-structure, and particle-sector applications are reserved for separate projection-level studies.

Keywords: finite register signature, block condensation, retention, physical readability, continuable difference, openness to continuation, recombinatorial structure, BBP-related limit form, foundations of physics

1 Introduction

A finite register signature can reproduce the coefficient and slot structure of the Bailey–Borwein–Plouffe series for π before any physical constant enters the construction. The signature comes from a single block-condensed register path,

$$P_G = 0 \mid 222 \mid 3 \mid 111,$$

whose non-zero slot vector $(1, 4, 5, 6)$ carries coefficients $(4, -2, -1, -1)$. Recursive level projection turns this into

$$\sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right),$$

the classical BBP form for π [1, 2].

Numerical proximity is not the point, and neither is physical identification. The construction asserts no empirical identity at this stage. The narrower question is this: Why should a finite block-condensed register lead to this recursive limit form without being fitted to it? And what has to hold for that result to count as more than coincidence?

Three findings carry the reconstruction. Block condensation turns a finite register path into a coefficient vector whose recursive projection has BBP form. That signature requires retention: an open block effect must remain structurally available beyond its own occurrence without displacing the core that carries it. The BBP-like path does not stand alone. It sits inside a complete finite execution space of 140 admissible paths, all generated by the one compact inventory 1112223.

Physical readability denotes the condition under which a formal structural role can be interpreted as a physical concept. A role becomes physically readable only after it has become stable, repeatable, and comparable within a formal apparatus. Space, time, causality, field-like structure, particle-like units, and mass-like behaviour are therefore not introduced as primitives. They enter only as projection readings of structural roles. The apparatus therefore does not construct physical reality. It reconstructs formal conditions under which already existing physical readability can be expressed without presupposing physical primitives.

The article extracts and sharpens the formal core of a broader German-language preprint, where the structural framework, the numerical signatures, and the extended projection vocabulary are developed in greater detail [3]. The argument here is deliberately narrower. It isolates four things: the register signature, retention, the finite path space, and the projection threshold at which physical readability begins.

The broader terminology is reduced here to the minimal continuability triad used in Section 5: self-binding $\mathfrak{S}$, difference order $\mathfrak{D}$, and openness to continuation $\mathfrak{O}$. These terms are not introduced as additional physical variables, but as reconstruction conditions that exclude cancellation, incompatibility, and rigid fixation.

The novelty is not the general claim that spacetime may be emergent. It lies in the intermediate formal layer placed between pure difference and physical interpretation. A finite register signature is treated not as an isolated numerical coincidence but as

the first visible trace of a wider execution space. Retention accounts for how open effects remain structurally available. The 140-path space shows that the signature belongs to the complete set of admissible executions of the same inventory. Inventory-preserving recombination turns this path space into a coupling apparatus in which stable roles can be defined. Physical readability begins only at this later threshold. The framework does not move directly from number to physics; it moves from register signature to retained path structure, from path structure to coupling roles, and only then to projection readings.

The contribution is concentrated by design: a register construction, a BBP-related internal limit signature, and a projection threshold for physical readability. Astrophysical residual regimes, flavour-sector constructions, fine-structure comparisons, and particle-sector applications are reserved for separate projection-level studies.

2 Finite register signature

2.1 Register form

The register form is not chosen because it fits the BBP series. It follows from the smallest relational structure in which simple reference positions, pair relations, and triadic mediation can all be represented together. Take three reference positions a, b, c . The combinatorial classes are

$$\emptyset, \quad a, b, c, \quad ab, ac, bc, \quad abc.$$

Grouped by relational order, they have the frequency structure

$$1, 3, 3, 1.$$

This is a multiplicity structure. It fixes how many roles of each relational level occur in the register:

$$\binom{3}{0} = 1, \quad \binom{3}{1} = 3, \quad \binom{3}{2} = 3, \quad \binom{3}{3} = 1.$$

It says nothing about the load of those roles. Multiplicity and load are distinct: multiplicity counts how many roles of a given level occur, while load evaluates the structural level of each role once it is there.

The register notation therefore translates the multiplicity structure into

$$R_3 = 0\ 111\ 222\ 3.$$

The leading 0 marks the initiating fixation of a register run. The 1-roles mark simple relational processing, the 2-roles mark pair formation, and the 3-role marks triadic mediation, or saturation. At this level the symbols are functional positions, not physical magnitudes.

Why triadic, and not smaller or larger? The dyadic form $R_2 = 0\,1\,1\,2$ can represent one pair relation but cannot mediate several at once. The tetradic form already demands invariance between overlapping triads. The triadic form is the first register in which several pair relations close within a single mediation:

$$ab, ac, bc \longrightarrow abc.$$

Once the register form is fixed, the inventory stays constant. The variable compact inventory is

$$I = \{1, 1, 1, 2, 2, 2, 3\}.$$

No valid operation may introduce, delete, or revalue a symbol. Inventory preservation is the operational form of structural resource constancy: continuation has to be carried within a fixed difference basis, not by adding further degrees of freedom after the fact.

2.2 Block condensation

The BBP-related signature comes from

$$P_G = 0 \mid 222 \mid 3 \mid 111.$$

This path is the globality pole: the saturated role 3 sits between two condensed role blocks. The term is descriptive; it carries no cosmological commitment.

The path is not read off by uniform symbol counting. It is read by block condensation. A block is a maximal connected run of equal roles, and its effect is anchored at one effective position. A 1-block acts at its beginning: repeated 1-roles are local repetitions of simple difference processing, and no pair-dependence has to be completed before the load is determinable. A 2-block acts at its end: repeated 2-roles form a bound pair-formation sequence whose cumulative load becomes invariantly determinable only at closure. The singular roles 0 and 3 act at their own positions. The result is an eight-slot vector signature.

The coefficients are fixed at a separate level. They do not follow from the multiplicities 1, 3, 3, 1 themselves. They follow from the role meaning inside the already fixed triadic register $R_3 = 0\,1\,1\,1\,2\,2\,2\,3$.

Within this register, the levels 1, 2, 3 do not merely name ranks. They indicate how many of the three jointly mediable difference layers are processed by the role. A 1-role processes one difference layer, a 2-role processes two coupled difference layers, and the 3-role marks the saturated mediation of all three difference layers. Full mediation therefore corresponds to three processed difference layers. A role of level $n \in \{1, 2, 3\}$ is evaluated by its share of this triadic saturation scale:

$$n \mapsto -\frac{n}{3}.$$

The sign marks that processed difference contributes load; the denominator marks the three-layer saturation scale of R_3 . Thus

$$1 \mapsto -\frac{1}{3}, \quad 2 \mapsto -\frac{2}{3}, \quad 3 \mapsto -1.$$

The compact inventory 1112223 then carries total processed difference load

$$3 \left(-\frac{1}{3} \right) + 3 \left(-\frac{2}{3} \right) - 1 = -4.$$

The initiating role 0 closes this register load:

$$0 \mapsto +4.$$

The full register inventory is therefore drift-free under the triadic saturation scale of R_3 . This gives the coefficient assignment

$$0 \mapsto 4, \quad 1 \mapsto -\frac{1}{3}, \quad 2 \mapsto -\frac{2}{3}, \quad 3 \mapsto -1.$$

Under block-condensed evaluation, the globality pole yields

$$\sigma_b(P_G) = (4, 0, 0, -2, -1, -1, 0, 0).$$

The non-zero entries occupy slots 1, 4, 5, 6, and the coefficient pattern is

$$(4, -2, -1, -1).$$

What matters about this vector is that it appears before any infinite series is written down. The recursive series does not create the coefficient pattern. It exposes a limit form already encoded in the block-condensed register vector.

The opposite fully condensed pole is

$$P_L = 0 \mid 111 \mid 3 \mid 222,$$

which yields

$$\sigma_b(P_L) = (4, -1, 0, 0, -1, 0, 0, -2),$$

and therefore

$$\Pi(\sigma_b(P_L)) = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{1}{8k+2} - \frac{1}{8k+5} - \frac{2}{8k+8} \right) \approx 3.0591043868.$$

This counter-pole is important. The same block-condensation convention does not send every extremal form to π : the BBP form is specific to the globality pole, while the locality pole carries a distinct residual form.

2.3 Recursive projection

The projection separates two counts. The visible register period has eight slots. The retention trace adds an address-parallel holding capacity for the same eight slots. A fully evaluated level is therefore carried by $8 + 8 = 16$ evaluation units, while the visible denominator positions stay at $8k + i$.

Let $\sigma = (\sigma_1, \dots, \sigma_8)$ be an eight-slot register vector. The recursive level projection is

$$\Pi(\sigma) = \sum_{k=0}^{\infty} \frac{1}{16^k} \sum_{i=1}^8 \frac{\sigma_i}{8k + i}.$$

The projection deliberately separates address period and level evaluation. The visible address period is eight, because the register has eight slots. The recursive level scale is sixteen, because a level is evaluated only when visible occurrence and retained availability are taken together. Hence the denominator address remains $8k + i$, while the level weight is 16^{-k} .

For $\sigma_b(P_G) = (4, 0, 0, -2, -1, -1, 0, 0)$ this gives

$$\Pi(\sigma_b(P_G)) = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right),$$

the BBP form for π .

The $8 + 8$ projection geometry is motivated by the split between visible register addressability and address-parallel retention. The first eight units are the visible register positions of the execution level. The second eight units do not introduce a second visible register and are not inserted to imitate the BBP scaling factor. They express the retention requirement of the apparatus: a block effect must remain addressably available after its local occurrence, otherwise block condensation would yield only a finite signature and not a continuable projection structure.

Thus a fully evaluated recursive level consists of visible execution plus retained address capacity. The denominator positions remain $8k + i$, because visible addressability is eight-periodic. The level factor 16^{-k} belongs to the complete evaluation unit, because each recursive level contains both the visible register trace and its retained counterpart. The $8 + 8 = 16$ geometry is therefore exposed, but not arbitrary: without the retained address layer, the open block effect would not remain structurally available across continuation.

The BBP pattern is an internal limit signature, not a derivation of π as a physical constant. The construction starts from a finite path, a block-condensation convention, and an eight-slot recursive projection, and the result is not a fitted approximation. Even so, coincidence with a known BBP series does not, on its own, identify the formal role with any physical magnitude. Under the stated block-condensation and recursive projection rules, the block-condensed signature becomes readable as a BBP-like internal limit form. This provides a non-trivial internal consistency and connection test for the structure, not a physical identity.

3 Retention and path space

3.1 Retention

Block condensation explains how a finite register path yields a concentrated vector signature. It does not yet explain how open effects stay available across further continuation. Retention supplies this condition.

In the narrow register sense, retention is not an additional operation beyond block condensation. It is the consequence of block condensation whenever a block load is not determinable at each local occurrence, but only at its effective block position. The 2-block makes this explicit: a 2-role denotes pair formation, consecutive 2-roles form a bound coupling sequence, and the cumulative block load becomes invariantly determinable only at block closure. The preceding 2-roles must therefore remain structurally available until the closing position is reached. This held availability is what is meant here by retention.

Retention is thus not a second register and not a hidden variable. It is the address-preserving consequence of the block rule itself: the block effect is carried until the position at which its load can be assigned invariantly.

In the broader structural sense, retention is the held availability of an open effect beyond its immediate occurrence, without replacement of the carrying core. It is neither psychological memory nor storage in a container. It is the formal persistence of availability. A retained effect can keep constraining later projection without being identical to the original block event.

Retention is the operational correlate of openness to continuation. Openness allows further relation, but without retention that relation would leave no held structural effect. Self-binding preserves a core, but without retention the core would carry no open consequences beyond itself.

Mass-like readability makes the need explicit. A mass-like role is not just a strongly bound role. It is a role whose open effects stay available and resist immediate dissolution:

$$\text{self-binding} + \text{retention} + \text{constrained openness} \rightsquigarrow \text{mass-like readability.}$$

This is a projection criterion, not a mass formula.

3.2 The complete path space

The compact inventory

$$I = \{1, 1, 1, 2, 2, 2, 3\}$$

has exactly

$$\frac{7!}{3!3!1!} = 140$$

admissible execution orders. The full path space is

$$\mathcal{P} = \{0||A \mid A \text{ is a permutation of } 1112223\}, \quad |\mathcal{P}| = 140.$$

The number 140 is not a symbolic figure. It is the cardinality of the complete admissible execution space of the compact inventory. No claim is made that the world consists of 140 objects. The claim is that a finite, complete execution space can serve as the first combinatorial path space of the theory.

The position of the saturation role 3 fixes an embedding axis. A path is bilaterally embedded if the 3-role lies at neither boundary of the variable part:

$$0 < \text{pos}_3(P) < 6.$$

If the 3-role sits at a boundary, the path is a boundary form. There are

$$\frac{6!}{3!3!} = 20$$

paths with 3 at the left boundary and the same number at the right. Hence the count of bilaterally embedded paths is

$$140 - 20 - 20 = 100.$$

The 100/40 split is not a physical count. It separates bilaterally embedded forms from boundary forms inside the fixed execution space.

4 Recombination and projection threshold

4.1 Inventory-preserving recombination

The 140-path space would remain a finite catalogue unless its paths could enter admissible relations. Recombination supplies that step: it turns admissible execution orders into a coupling space in which structural roles can become recurrent. For two compact path parts A and B , ordered cut plans I and J divide both sequences into segments. Segmental recombination reassigns these already ordered segments:

$$T_{I,J}(A, B) = (C, D).$$

The operation is admissible only if it preserves the fixed inventory and returns both outputs to the same path space:

$$0 \parallel C \in \mathcal{P}, \quad 0 \parallel D \in \mathcal{P}.$$

If that fails, the recombination is formally constructible but structurally non-admissible.

Two coupling layers are distinguished. Core coupling tests self-connection under bilateral segment participation: both paths contribute a non-empty segment at every effective assignment position. Asynchronous support coupling allows one-sided holding or continuation: at least one segment is non-empty, while the other path may

contribute an empty holding segment. Thus, for the same cut complexity,

$$S_{\text{core}}(k) \subset S_{\text{async}}(k).$$

The difference is not temporal. It is structural: core coupling tests self-connection, asynchronous coupling tests supported connection.

4.2 Pair-coupling domain

Recombination concerns path pairs, not isolated paths. The formal coupling domain is therefore the set of unordered path pairs, self-pairs included:

$$\mathcal{P}^{(2)} = \{\{P, Q\} \mid P, Q \in \mathcal{P}\}, \quad |\mathcal{P}^{(2)}| = \frac{140 \cdot 141}{2} = 9870.$$

The number 9870 is used only to fix the complete finite domain on which recombinational recurrence can be tested. It is not interpreted numerologically and has no physical reading by itself. Its role is to prevent the coupling construction from resting on selected examples: every admissible unordered pair of paths has a formally defined location in $\mathcal{P}^{(2)}$.

A location in $\mathcal{P}^{(2)}$ is not a physical object. It is a valid pair-location in the formal coupling domain. An effective edge appears only when an admissible recombination maps one valid pair-location to a different one; identity recombinations are ignored.

The detailed M_0 – M_3 status matrix is developed in the long version [3]. The present article uses only its underlying distinction: bilaterally embedded forms differ from boundary forms, and core self-connection differs from asynchronous support connection. This distinction is where the finite path catalogue becomes structurally expressive: it separates forms that carry their own embedding from forms that remain supported, virtual, or open.

The aim of the coupling construction is not to classify every path-pair region in the present article, but to define the closed finite domain in which recurrence, support, and role formation can be tested without leaving the fixed inventory. In this sense, the pair domain is a methodological control: the internal limit signature is not placed beside an informal set of examples, but inside a finite recombinatorial apparatus whose possible pair-locations are completely specified.

4.3 Projection threshold

A structural role is a formally determined role in the path and coupling space that remains identifiable across admissible recombinations. A structural mode is a classified structural role—classified, for instance, by register load, block condensation, effect position, retention, recurrence under recombination, and connection capacity. A projection class is a structural mode that is repeatably and comparably readable as a physical notion. A physical reading connects such a class to terms such as space, time, causality, field, particle-like unit, or mass-like behaviour.

The four levels must remain distinct:

structural role $\rightarrow$ structural mode $\rightarrow$ projection class $\rightarrow$ physical reading.

This separation prevents two illegitimate leaps at once: the leap from formal structure straight to physical ontology, and the leap from numerical proximity straight to empirical identification. It also marks the point at which the construction becomes internally coherent. The BBP-related expression is not the endpoint of the theory. It is the first case in which register load, block condensation, retention, recursive projection, finite path multiplicity, and coupling structure all become mutually constrained.

5 Continuability and physical readability

5.1 Continuability conditions

The construction so far needs a general account of continuability. A difference is not yet an object, a field value, an event, a position, or a relation between already constituted terms. Most possible differences leave no structural remainder. A difference becomes relevant only where it is not fully cancelled, and where that non-cancellation can be carried forward without immediate fixation or incompatibility.

Three limiting failures must therefore be excluded: cancellation, incompatibility, and rigid fixation.

Proposition 1 (Continuability condition) *A configuration can become structurally, and later physically, readable only if it avoids cancellation, incompatibility, and rigid fixation.*

The corresponding reconstruction conditions are self-binding, difference order, and openness to continuation:

$$\mathcal{C} = (\mathfrak{S}, \mathfrak{D}, \mathfrak{O}).$$

Self-binding $\mathfrak{S}$ is the condition by which a configuration refers back to the conditions of its own persistence. Difference order $\mathfrak{D}$ is the condition by which persisting differences remain mutually compatible without cancelling each other. Openness to continuation $\mathfrak{O}$ is the condition by which a configuration remains non-closed and capable of further relation without dissolving at once.

The notation

$$\mathfrak{S} \neq 0, \quad \mathfrak{D} \neq 0, \quad \mathfrak{O} \neq 0$$

is not metric. It marks the exclusion of the three limiting failures. Without $\mathfrak{S}$, differences cancel. Without $\mathfrak{D}$, they cannot coexist. Without $\mathfrak{O}$, they become fixed and cannot continue.

5.2 Physical readability

Physical concepts become readable where formal structural roles stabilize into repeatable projection classes. Space, time, causality, field-like structure, particle-like units,

and mass-like behaviour are not primitives of the reconstruction. They are projection readings.

Space is read as coexistence order. A spatial reading becomes possible when several structural roles can be maintained together without cancelling one another. Time is read as directed continuation order: a temporal reading becomes possible where simultaneous consistency cannot be maintained and continuation has to be executed in sequence. Causality is read as continuation dependence—a causal reading becomes possible where the admissibility of a later role depends on a prior structural embedding.

Field-like readings arise where extended regions of the coupling graph preserve recognizable connection patterns. Particle-like readings arise where a structural role recurs as a stable unit under repeated recombination. Mass-like readings arise where self-binding, recurrent retention, and constrained openness jointly make a role resistant to immediate dissolution. These are projection criteria, not completed physical derivations.

6 Relation to existing approaches and scope

The framework shares with causal set theory, causal dynamical triangulations, loop quantum gravity, spin-network approaches, and holographic or information-theoretic approaches the refusal to treat classical spacetime as simply given [4–11]. The starting point is what differs. It does not begin with causal order, triangulated histories, spin-network states, quantum information, entanglement entropy, or algebraic observables. It begins with finite register signatures and the structural continuability of difference.

In relation to these approaches, the distinguishing feature is neither discreteness alone nor emergent spacetime in general; several foundational programs share both. The specific contribution is a combination: a triadic register with multiplicity/load separation, block-condensed effect positions, a retention trace, two extremal poles with different limit behaviour, a complete finite path space, and an inventory-preserving recombination apparatus that defines a projection threshold for physical readability.

The difference has consequences. If causal order is primitive, a temporal or quasi-temporal relation has already entered the reconstruction. If quantum information is primitive, a specific formal theory is already presupposed. If entanglement is primitive, the framework starts from a quantum state structure. The present construction is more austere: it asks how any physical readability can arise from constrained continuability before the usual physical categories are admitted at all.

The present article separates the core construction from the broader projection program. Extended applications—field-like coupling signatures, flavour-sector constructions, lepton-like and neutrino-like mass structures, astrophysical residual regimes, and further numerical connection tests—belong to that broader program and require separate operational maps. Parts of this broader program are discussed in the long version [3]. The scope here is narrower: it fixes the register apparatus, the BBP-related internal limit signature, retention, the 140-path execution space, and the conditions under which formal roles become physically readable.

The established result is therefore not merely the BBP-related expression. It is the controlled sequence—from finite register signature to retention, complete path

space, admissible recombination, and projection threshold. That sequence is the formal contribution.

Two projection choices require explicit status control. First, the linear load rule $n \mapsto -n/3$ fixes the register coefficients and yields a drift-free balance over the compact inventory. Its role is not that of an empirical fit. It expresses the triadic saturation scale inside the already fixed register R_3 : a level- n role processes n of the three difference layers that can be jointly mediated within this register. The resulting balance is therefore internal to the stated register apparatus. Second, the $8 + 8 = 16$ projection geometry follows the split between visible register positions and address-parallel retention trace, and its role is to define the recursive projection convention used for the internal limit signatures. Both belong to the current projection apparatus, and both delimit what the present argument establishes. The BBP correspondence is read as an internal limit signature of the block-condensed register projection, not as a physical identity. In the same way, the path space, coupling graph, and status classes define structural roles and projection candidates; they do not yet constitute a field dynamics, a particle spectrum, or an empirical theory of constants.

The next step is not a further justification of the BBP expression. It is an arithmetic of configuration parameters capable of distinguishing loop depth, retained effect structure, coupling support, and openness channels across different projection classes.

7 Conclusion

Physical readability can be reconstructed from a coherent chain of internal structural findings: a BBP-related block-condensed register signature, retention as the held availability of open structural effect, the complete 140-path execution space generated by the compact inventory 1112223, and an inventory-preserving coupling apparatus in which structural roles can become recurrent.

These findings point toward a deeper reconstruction. From the exclusion of cancellation, incompatibility, and rigid fixation, three minimal conditions of continuability follow: self-binding, difference order, and openness to continuation. They support a register form, a finite path space, and an inventory-preserving recombination structure on unordered path pairs.

The resulting coupling graph does not describe the motion of objects. It describes the formal embedding conditions under which stable structural roles can arise. Space, time, and causality enter only as projection readings—space as coexistence order, time as directed continuation order, causality as continuation dependence. Field-like, particle-like, and mass-like readings are treated as later projection classes of structural modes.

The result is a foundational construction, not an empirical physical theory. It reconstructs the threshold at which physical description becomes possible. Its core claim is narrow and definite: physical notions become readable from stable roles in a finite, retained, recursively extendable recombinatorial structure.

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Data availability

No empirical datasets were generated or analysed in this article. The formal constructions described in the manuscript are reproducible from the definitions and operations given in the text.

Author contributions

The author is solely responsible for the conceptual development, formal reconstruction, and manuscript preparation.

Use of generative AI and AI-assisted technologies

During the preparation of this work, generative AI and AI-assisted tools were used for editorial support, structural consolidation, language refinement, consistency checks, and exploratory formulation of formal representations. The author reviewed, selected, revised, or rejected all such outputs and takes full responsibility for the content of the manuscript.

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